

When Do Physics Priors Help?

A Controlled Benchmark of Physics-Informed Dynamics Models for Underactuated Swing-Up

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Abstract—Physics-informed machine learning promises data-efficient, physically consistent dynamics models, but it is often unclear *when* a physics prior pays off relative to a black-box baseline—or whether a mismatched prior can hurt. We present a controlled, reproducible benchmark of seven learned dynamics models—a multilayer perceptron (MLP), a polynomial system-identification baseline, a Gaussian process (GP), a Lagrangian Neural Network (LNN), a naive Hamiltonian Neural Network (HNN), a structured HNN with a learned mass matrix, and a Lagrangian-informed GP (LGP)—on two underactuated systems, the Acrobot and the Pendubot. We evaluate four axes: one-step accuracy, long-horizon rollout stability (valid prediction time), energy drift, and downstream energy-based swing-up control, across five seeds, four data budgets, and both clean and sensor/process-noise conditions. We find a nuanced map rather than a single winner. The LNN, whose learned positive-definite mass matrix matches these systems, is the most accurate and energy-consistent and never diverges. The GP is the most data-efficient and yields the most stable rollouts despite modest one-step accuracy. The LGP blends the two, achieving LNN-level accuracy with GP-level rollout stability and the lowest energy drift. The naive HNN, whose identity-mass assumption is violated by configuration-dependent inertia, is worse than the black-box MLP and diverges in 20–32% of rollouts; a controlled ablation shows that giving the Hamiltonian model a learned mass matrix—which by Legendre duality recovers the LNN—fully removes this failure, isolating the assumption rather than the formulation as the cause. Under noise all models degrade gracefully, but the GP residual in the LGP overfits noise and loses rollout stability. Only models with an energy function can drive energy-based control. The result is a concrete map of when physics priors help: scarce data, long horizons and energy-sensitive tasks, and a prior whose structure matches the system.

Index Terms—physics-informed machine learning, learned dynamics, underactuated robots, Lagrangian neural networks, Gaussian processes, swing-up control.

I. INTRODUCTION

Learning the dynamics of a mechanical system from data underpins model-based control and reinforcement learning. Plain neural networks fit one-step transitions accurately, yet rolled out over long horizons they drift and violate invariants such as energy conservation. Physics-informed machine learning (PiML) bakes structure—Lagrangian or Hamiltonian

mechanics, positive-definite inertia, energy conservation—into the model [1], [2], promising better long-horizon behaviour and sample efficiency.

In practice it is rarely obvious *when* a prior is worth its complexity, or whether a prior whose assumptions are violated by the real system can hurt. Most papers introduce one physics-informed model and show it beats a baseline on a favourable task; what is missing is a controlled, like-for-like comparison across priors, evaluation criteria, data budgets, and noise conditions.

We provide such a comparison and summarise it as a map of when physics priors help, with four findings:

- 1) **Matching structure matters most.** The LNN, which learns a positive-definite, configuration-dependent mass matrix, is the most accurate and energy-consistent model and never diverges.
- 2) **A mismatched prior is worse than none.** The naive HNN, assuming an identity mass matrix, is worse than the black-box MLP and diverges in 20–32% of rollouts. A controlled ablation pinpoints the assumption: giving the Hamiltonian model a learned mass matrix removes the failure.
- 3) **Uncertainty buys stability and data efficiency.** The GP is the most data-efficient model and yields the most stable rollouts; combining it with Lagrangian structure (the LGP) gives the best overall blend.
- 4) **Robustness has costs.** Under noise the GP is the most robust, while the LGP’s GP residual overfits noise and loses rollout stability.

All experiments are reproducible from a released, configuration-driven pipeline.

II. RELATED WORK

Physics-informed neural dynamics. Deep Lagrangian Networks (DeLaN) [3] learn the mass matrix and potential of a Lagrangian and recover the equations of motion by automatic differentiation; Lagrangian Neural Networks [4] generalise the idea. Hamiltonian Neural Networks [5] learn a Hamiltonian and integrate the symplectic flow, conserving energy but

requiring canonical coordinates; symplectic and variational-integrator variants extend them to control and rigid bodies [6], [7]. Physics-informed neural networks more broadly embed governing equations into the loss [2], [1].

Gaussian-process and classical dynamics. GPs provide calibrated uncertainty and data efficiency for model learning [8], [9] and underpin sample-efficient control such as PILCO [10]. Sparse regression on a feature library (SINDy-style) is a classical system-ID baseline [11]. Hybrid schemes learn a data-driven residual on top of a parametric or structured model.

Underactuated swing-up. The Acrobot and Pendubot are standard benchmarks; energy-shaping controllers swing them toward upright by regulating total energy, typically with an LQR catch for final balancing [12], [13]. We use swing-up as a downstream task that stresses the *usefulness* of a learned model, not only its predictive error.

We hold data, integrator, training objective, and evaluation fixed across all models, enabling a fair comparison rather than a single favourable demonstration.

III. SYSTEMS AND PROBLEM SETUP

We study two planar double pendulums with a single actuator, sharing the manipulator equation

$$M(q) \ddot{q} + C(q, \dot{q}) \dot{q} + g(q) = B u, \quad (1)$$

with joint angles $q = (q_1, q_2)$, mass matrix $M(q) \succ 0$, Coriolis term C , gravity $g(q) = \partial V / \partial q$, and actuation matrix B . The **Acrobot** is actuated only at the elbow ($B = [0, 1]^\top$) and the **Pendubot** only at the shoulder ($B = [1, 0]^\top$); the other joint is passive. The state is $x = (q_1, q_2, \dot{q}_1, \dot{q}_2)$ and u is a scalar torque; we integrate with RK4 at $\Delta t = 10^{-2}$ s. Given transitions (x_t, u_t, x_{t+1}) under random torques, the task is to learn a model that predicts accelerations, hence the next state via the same integrator. A single integrator and one-step objective are used for every model, so behavioural differences are attributable to the prior.

IV. THE MODEL ZOO

All models map (x, u) to a state derivative and are rolled out with RK4.

Vanilla MLP (black box). Regresses joint accelerations directly; no physical structure.

Polynomial (system ID). Closed-form ridge regression of accelerations on a degree-2 polynomial expansion of $[\sin q_1, \cos q_1, \sin q_2, \cos q_2, \dot{q}_1, \dot{q}_2, u]$; a cheap classical baseline.

Gaussian Process (GP). An exact GP with an ARD-RBF kernel regresses each acceleration component from the same features, fit by marginal likelihood; the $(\sin, \cos)$ encoding respects periodicity. We cap the training set at 1000 points for tractability.

Lagrangian Neural Network (LNN). Two networks output a Cholesky factor of $M(q) = L(q)L(q)^\top + \epsilon I \succ 0$ and a potential $V(q)$; the Coriolis and gravity terms follow by automatic differentiation and the acceleration from $M\ddot{q} = Bu - C\dot{q} - g$. The known B is supplied as a prior.

Naive HNN. A scalar Hamiltonian $H(q, p)$ with the identity-mass convention $p := \dot{q}$, integrated along the symplectic flow $\dot{q} = \partial H / \partial p$, $\dot{p} = -\partial H / \partial q + Bu$. It does *not* learn a configuration-dependent mass matrix.

Structured HNN. A Hamiltonian model that *learns* $M(q)$ and $V(q)$, with canonical momentum $p = M(q)\dot{q}$. By the Legendre transform its equations of motion coincide with the LNN's; we include it as a controlled ablation of the naive HNN's identity-mass assumption.

Lagrangian-informed GP (LGP). A hybrid: an LNN supplies a physics mean and a GP models the residual acceleration, $\ddot{q} = \ddot{q}_{\text{LNN}} + \text{GP}_{\text{res}}$, keeping structure while letting the GP correct what the structural model misses.

The structured networks conserve their own learned energy to $\sim 10^{-7}$ over 200 steps before training, confirming the autograd equations of motion.

V. EXPERIMENTAL SETUP

Data. Per system we generate 8000 transitions from random rollouts; a 20% tail is held out for one-step evaluation and the pool is subsampled to budgets of 250, 1000, 4000 to probe data efficiency. The noisy condition adds process noise (a random state kick each step) and sensor noise (added to recorded states), both with std 0.02.

Training. Each model minimises a one-step angle-aware MSE between the RK4-integrated prediction and the recorded next state (GP-style models are fit by marginal likelihood). We use 100 epochs, Adam, gradient clipping, and *five* seeds per configuration.

Axes. (1) one-step MSE; (2) *valid prediction time*, the steps a free-swing rollout tracks the truth before the per-step error exceeds a threshold (robust on chaotic systems, where raw long-horizon MSE saturates); (3) energy drift over the rollout, plus the fraction of rollouts that diverge; (4) swing-up control (Section VI-D). Rollouts use identical fixed initial conditions across models.

VI. RESULTS

A. Prediction, Stability, and Energy (Axes 1–3)

Table I reports the clean condition at the largest budget; Fig. 1 shows the data-budget sweep.

Accuracy and energy. The LNN is the most accurate model on both systems and among the lowest in energy drift. The Polynomial baseline is weakest, with large drift and 16–36% divergence. The MLP is a competent middle performer.

Stability. On the chaotic free-swing task the GP gives by far the longest valid prediction time ($> 2.5\times$ the MLP) despite larger one-step error: GP smoothing damps the error growth that destabilises the MLP. The LGP combines LNN-level accuracy with high valid time and the lowest energy drift, making it the best all-round model on the dynamics axes.

Data efficiency (Fig. 1). At the smallest budget the GP and Lagrangian models are already far more accurate than the MLP, whose advantage only appears with more data; physics priors and GP smoothing help most when data are scarce.

TABLE I

DYNAMICS AXES (CLEAN, $n=4000$, MEAN OVER FIVE SEEDS). ARROWS MARK LOWER ($\downarrow$) / HIGHER ($\uparrow$) IS BETTER. “DIV.” IS THE FRACTION OF ROLLOUTS THAT DIVERGED. THE STRUCTURED HNN COINCIDES WITH THE LNN (LEGENDRE DUALITY) AND IS ANALYSED SEPARATELY IN TABLE II.

System	Model	One-step MSE $\downarrow$	Valid steps $\uparrow$	Energy drift $\downarrow$	Div.
Acrobot	Vanilla MLP	6.1×10^{-3}	25.8	1.98×10^1	0%
	Polynomial	4.2×10^{-2}	13.2	6.7×10^2	36%
	Vanilla GP	8.5×10^{-3}	67.6	9.7	0%
	Lagrangian NN	1.4×10^{-3}	27.3	7.7	0%
	Naive HNN	1.6×10^{-2}	10.7	9.6×10^2	20%
	Lagrangian GP	1.5×10^{-3}	52.9	5.7	0%
Pendubot	Vanilla MLP	7.2×10^{-3}	28.0	1.87×10^1	0%
	Polynomial	4.1×10^{-2}	11.9	3.0×10^2	16%
	Vanilla GP	8.8×10^{-3}	71.4	1.13×10^1	0%
	Lagrangian NN	2.6×10^{-3}	26.9	9.9	0%
	Naive HNN	1.6×10^{-2}	12.2	1.0×10^3	32%
	Lagrangian GP	2.3×10^{-3}	58.6	6.7	0%

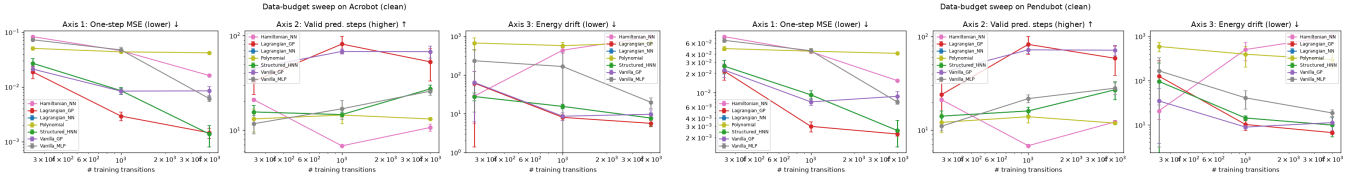


Fig. 1. Data-budget sweep (clean), Acrobot (left) and Pendubot (right): each dynamics axis versus the number of training transitions (log-log), mean over five seeds with standard-deviation error bars.

TABLE II

HNN ABLATION ON THE ACROBOT ($n=4000$, FIVE SEEDS). LEARNING THE MASS MATRIX REMOVES THE NAIVE HNN’S FAILURE; THE STRUCTURED HNN MATCHES THE LNN.

Model	Valid steps $\uparrow$	Energy drift $\downarrow$	Div.
Naive HNN (identity mass)	10.7	9.6×10^2	20%
Structured HNN (learned mass)	27.3	7.7	0%
Lagrangian NN (reference)	27.3	7.7	0%

TABLE III

NOISE ROBUSTNESS ON THE ACROBOT ($n=4000$): ONE-STEP MSE AND VALID STEPS, CLEAN $\rightarrow$ NOISY.

Model	One-step MSE (clean $\rightarrow$ noisy)	Valid steps (clean $\rightarrow$ noisy)
Vanilla MLP	$6.1 \times 10^{-3} \rightarrow 1.0 \times 10^{-2}$	25.8 $\rightarrow$ 29.6
Vanilla GP	$8.5 \times 10^{-3} \rightarrow 1.3 \times 10^{-2}$	67.6 $\rightarrow$ 59.3
Lagrangian NN	$1.4 \times 10^{-3} \rightarrow 3.3 \times 10^{-3}$	27.3 $\rightarrow$ 35.4
Lagrangian GP	$1.5 \times 10^{-3} \rightarrow 3.3 \times 10^{-3}$	52.9 $\rightarrow$ 36.4

B. The HNN Failure Is the Assumption, Not the Formulation

The naive HNN is the worst structured model: poor one-step accuracy, the fewest valid steps, and 20–32% divergence (its huge drift values are an artifact of these divergences). Table II isolates the cause. Replacing the identity-mass assumption with a learned mass matrix—the Structured HNN, which by Legendre duality has the same equations of motion as the LNN—removes the failure entirely: divergence drops to 0% and valid steps and drift match the LNN. The naive HNN’s failure is therefore due to the violated identity-mass assumption, not to the Hamiltonian formulation.

C. Robustness to Noise

Table III reports the clean $\rightarrow$ noisy change at $n=4000$. All models degrade gracefully (one-step error roughly doubles). Two effects stand out: the GP remains the most stable in absolute terms, while the LGP’s GP residual *overfits* the noise and loses rollout stability (valid steps fall from 52.9 to 36.4 on the Acrobot), eroding its clean-condition advantage. Hard structural priors (LNN) are comparatively noise-tolerant.

D. Downstream Swing-Up Control (Axis 4)

We drive an energy-shaping controller on the *true* plant using a model’s energy estimate,

$$u = k_e (E_{\text{top}} - E_{\text{model}}(x)) \dot{q}_{\text{act}} - k_p q_{\text{act}} - k_d \dot{q}_{\text{act}}, \quad (2)$$

clipped to a torque limit, with E_{top} the upright energy under the same model. Only models with an energy function (LNN, Structured HNN, Naive HNN, LGP) can drive it; the MLP, Polynomial and GP cannot, which is itself a result. We report the mean normalised tip height reached (0 down, 1 upright)—a continuous, robust measure of swing-up quality—and a success rate, with a true-energy Oracle as the ceiling (Table IV, Fig. 2).

Models with an accurate energy (LNN, LGP, Structured HNN) swing the tip to $\sim 65\%$ of full height, well above the naive HNN; black-box models cannot attempt the task. No learned model matches the Oracle, and robustly *balancing* the underactuated systems at the top—which requires an LQR catch fed by precise swing-up entry—remains hard; we therefore treat the continuous tip-height as the primary control

TABLE IV
ENERGY-BASED SWING-UP (FIVE SEEDS). MEAN TIP HEIGHT AND SUCCESS RATE; THE ORACLE USES THE TRUE ENERGY. BLACK-BOX MODELS HAVE NO ENERGY FUNCTION (N/A). THE STRUCTURED HNN MATCHES THE LNN.

System	Model	Mean tip height	Success
Acrobot	Oracle	0.97	100%
	Lagrangian NN	0.63	25%
	Lagrangian GP	0.63	25%
	Naive HNN	0.51	0%
Pendubot	Oracle	0.96	100%
	Lagrangian NN	0.65	15%
	Lagrangian GP	0.65	15%
	Naive HNN	0.66	10%

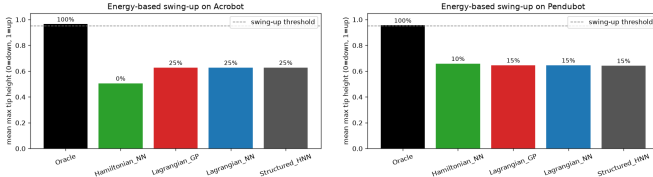


Fig. 2. Swing-up: bars are mean tip height, percentages are success rate. Black-box models are absent (no energy function).

metric and provide an optional LQR catch in the released code. The ranking nonetheless tracks energy-model quality, connecting predictive accuracy to downstream usefulness.

VII. DISCUSSION: A MAP OF WHEN PHYSICS PRIORS HELP

Across the four axes a consistent picture emerges. *First*, physics priors help most when their structure matches the system: the LNN’s learned positive-definite, configuration-dependent mass matrix is the single most valuable ingredient, and the Structured-HNN ablation shows the same structure recovers a failing model. *Second*, a mismatched prior is actively harmful—the naive HNN’s identity-mass assumption makes it worse than a black box and unstable. *Third*, the benefit of any prior is largest with scarce data and on long-horizon, energy-sensitive tasks; at high data with only one-step accuracy in view, the MLP is competitive. *Fourth*, uncertainty-aware models (GPs) buy rollout stability and data efficiency complementary to hard priors, and combining them (the LGP) is best on clean dynamics—but its residual overfits noise, a robustness cost. No single model wins everything, which is the practical guidance a practitioner needs.

VIII. LIMITATIONS AND FUTURE WORK

Our study is in simulation on two planar systems. Absolute swing-up success is modest because robust balancing of underactuated systems is a hard control problem in its own right; per-model model-predictive control would also let black-box models be evaluated downstream. The exact GP is capped in training-set size, and is near-deterministic at the largest budget. Extensions include additional systems (e.g. cart-pole), real-

robot data, larger noise sweeps, and sparse or deep-kernel GPs to scale the GP and LGP.

IX. CONCLUSION

We presented a controlled, reproducible benchmark of seven dynamics models on two underactuated systems across four axes, five seeds, and clean and noisy conditions. Rather than crowning a method, our results map out *when* physics priors help: most when the prior matches the system’s structure, when data are scarce, and on long-horizon, energy-sensitive tasks—while a mismatched prior can be worse than none. We release the pipeline as a reusable testbed.

REPRODUCIBILITY

Code, configurations, datasets, and the scripts that regenerate every table and figure are available at <https://github.com/amirhossein-fattahi/piml-underactuated>.

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